A Survey of Image Transformation Based Network Intrusion Detection Systems*

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Abstract

AI-driven cyberattacks demand accurate intrusion detection systems (IDS). Recent studies transform network traffic into two-dimensional images to enable the use of vision models. This paper reviews four Transformer-based image-conversion IDS studies and compares them by conversion unit, conversion method, temporal information, and intent of conversion.

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1 Introduction

Modern network traffic exhibits complex structural and temporal characteristics that challenge deep-learning-based IDS, and detection performance depends on both classifier design and input representation. Image-based IDS transforms traffic into 2D images so that vision models can learn feature relationships difficult to express in 1D vectors [1] [2] [3] [4]. Such approaches are relevant to trustworthy e-business information systems, where resilient network infrastructures are essential.

2 Image-Based IDS Research

Ho et al. [1] map flow features (selected via a Decision Tree) to RGB images, and use overlapping sliding windows to capture correlations among adjacent flows. The images are classified by a ViT.

DE-ViT [2] reshapes each 64-dimensional flow vector into an 8×8 feature image and applies Deformable Attention with Layered Focal Loss to mitigate class imbalance.

ConViTML [3] maps 784 packet bytes to a 28×28 grayscale image. ConViT-based meta-learning supports new attack adaptation, while positional embedding preserves session order.

ImagTIDS [4] encodes sequential flow features into images via Gramian Angular Difference Field (GADF) and classifies them with ImagTrans, a CNN-Transformer hybrid.

3 Comparison

Table 1 shows that each method preserves different information through image conversion, making the conversion strategy a key IDS design factor.

Aspect	Ho et al. [1]	DE-ViT [2]	ConViTML [3]	ImagTIDS [4]
Conversion unit	Multiple flows	Single flow	Single packet	Sequential flow features
Conversion method	Linear RGB mapping	2D reshape of feature vector	2D reshape of packet bytes	GADF encoding of time series
Temporal information	Window overlap	Not explicit	Packet order	Direct encoding
Intent of conversion	Capture inter-flow relations	Enable ViT on feature vectors	Enable ViT on packet bytes	Encode inter-timestep relations

However, none of the surveyed studies analyze how image conversion contributes to detection performance, and the use of different evaluation datasets across studies makes it difficult to attribute the reported gains to the image conversion alone.

4 Conclusion

Image-transformation-based IDS approaches enable vision models to learn structural or temporal traffic relationships that 1D vectors cannot easily capture. Future work should select conversion strategies according to detection objectives, deployment environments, and traffic characteristics.

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References

[1] Ho, C. M. K., Yow, K.-C., Zhu, Z., & Aravamuthan, S. (2022). Network intrusion detection via flow-to-image conversion and vision transformer classification. *IEEE Access*, *10*, 97780–97793. https://doi.org/10.1109/ACCESS.2022.3200034

[2] He, K., Zhang, W., Zong, X., & Lian, L. (2024). Network intrusion detection based on feature image and deformable vision transformer classification. *IEEE Access*, *12*, 44335–44350. https://doi.org/10.1109/ACCESS.2024.3376434

[3] Yang, L., Guo, S., Liu, D., Zeng, Y., Jiao, X., & Zhou, Y. (2024). ConViTML: A convolutional vision transformer-based meta-learning framework for real-time edge network traffic classification. *IEEE Transactions on Network and Service Management*, *21*(3), 3344–3357. https://doi.org/10.1109/TNSM.2024.3383218

[4] Wang, P., Song, Y., Wang, X., Guo, X., & Xiang, Q. (2025). ImagTIDS: An internet of things intrusion detection framework utilizing GADF imaging encoding and improved transformer. *Complex & Intelligent Systems*, *11*, Article 93. https://doi.org/10.1007/s40747-024-01712-9